Quadratic equations

1 $x = -12$

2

a $x = \pm\sqrt{2}$ **b** $x = \pm 5$ **c** $x = \pm 11$ **d** $x = \pm 7\sqrt{6}$

e $x = \pm 11$ **f** $x = \pm 5$ **g** $x = \pm 8$ **h** $x = \pm 15$

i $x = 4\,, -10$ **j** $x = 11\,, -5$ **k** $x = 6 \pm \sqrt{2}$ **l** $x = -7\,, 11$

m $x = 16\,, -2$ **n** $x = -2\,, 16$ **o** $x = \frac{7}{8}\,, \frac{25}{8}$ **p** $x = \pm\frac{4}{9}$

3

a $x = 0\,, -7$ **b** $x = 0\,, \frac{9}{2}$ **c** $x = \frac{9}{10}$ **d** $x = \frac{9}{4}$

e $x = \frac{3}{7}$ **f** $x = 4\,, 2$ **g** $x = 6\,, -7$ **h** $x = \frac{5}{8}\,, \frac{7}{3}$

i $x = -\frac{8}{3}\,, \frac{7}{5}$ **j** $x = \frac{17}{3}\,, \frac{d}{2}$

4

a $y = \pm 5$ **b** $y = \pm\frac{6}{5}$ **c** $k = \pm 2$ **d** $k = \pm\frac{4}{9}$

e $v = \pm\frac{8}{5}$ **f** $p = \pm 10$ **g** $m = 0\,, -5$ **h** $m = 0\,, -5$

5 $x = \pm\sqrt{\frac{c}{a}}$

Applications

6 $k = -16$

7 $a = 13$

8 $t = 14,\ t = 10$

9

a $x = 0, 13$

b $13\,\text{cm}$

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